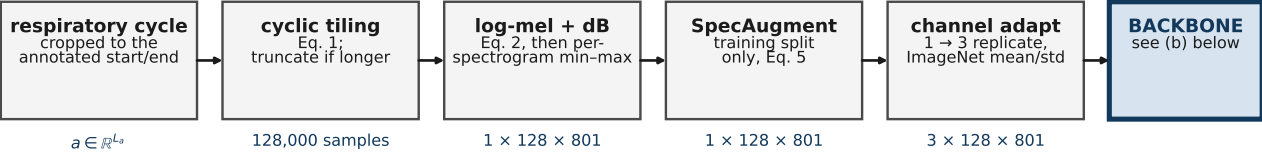


(a) Shared front end — identical for all four backbones



(b) The four interchangeable backbones — one recipe, one seed, one preprocessing cache

backbone	family	parameters	GAP input	ICBHI
MobileNetV2	CNN, inverted res.	2,228,996	$1280 \times 4 \times 26$	0.5602
Swin-T	shifted-window attn.	27,522,430	768 (pooled)	0.5304
DeiT-S	ViT + distill. token	21,667,204	384 (CLS)	0.5149
ViT-B/16	plain ViT	85,801,732	768 (CLS)	0.4994

MobileNetV2 path, in detail
conv 3×3 , stride 2 $\rightarrow$ 32 channels
17 inverted residual bottlenecks (expansion 6)
conv $1 \times 1 \rightarrow$ 1280 channels
total stride 32 $\rightarrow$ $1280 \times 4 \times 26$

Grad-CAM tap
read at the 4×26 grid above, so one cell is $\approx 500 \text{ Hz} \times 0.31 \text{ s}$; every attribution map is that grid bilinearly upsampled, not fine structure

The 2.23 M-parameter convolutional backbone beats all three transformers under one shared recipe; ViT-B/16 collapsed to the majority class in both of its runs.

(c) Shared classifier head and objective — also identical for all four

